

7 Steps to your SOC Analyst Career



How to begin your career as a SOC Analyst in 2024

Career UP SOC Analyst Academy
Learn more by visiting:
<https://www.careerup.tech>

My introduction:

Hi, I am Chris and I am a recognized Cybersecurity professional with over 20 years experience in private and public spaces. During my career, I have authored courses and mentored hundreds candidates from Entry to Senior levels to establish and enhance their careers.

In my career, I have held positions as a Senior Security Engineer, Senior Security Analyst, Malware Analyst, Lead Forensics Investigator, Cybersecurity Engineering Manager, Senior Network Engineer, Principle Network Engineer, CTO, and Director.

Throughout my career, I have been fortunate to work with some of the top engineers in the world. They helped mentor me and in return, I have helped and mentored hundreds others from entry-level with no experience to senior level to help them further their careers.

Now, I am providing you with this guide to help you begin your SOC Analyst career in 2024.

How to use this e-book:

Everyone is busy and time is the most valuable commodity in today's fast-paced digital world.

My goal when writing this e-book was to provide the reader with KEY information without requiring you to read hundreds of pages.

This free e-book provides a step-by-step guide that will provide you with the direction you need to begin a career as a SOC Analyst. It's up to you to take what is in this e-book and make YOUR plan and carry it through.

If there is one piece of advice that I could share with you to help you become successful, it is simply to:

1. Take action - Take the steps, perform the actions, put in the work that will pay off later on.
2. Never give up! You will have challenges, the truth is, everyone does. If you continue moving forward, you will reach your goals.
3. As of right now, if you are reading this, promise yourself that you will never doubt yourself OR let anyone make you doubt yourself.

I have helped hundreds begin their career and every single person who has been successful had something in common. They put in the time and effort, never gave up, and never backed down from a challenge, now it is your turn. Let's go!

As you begin, it is important to realize that everyone's path into becoming a SOC Analyst may differ.

There are many different ways to enter Cybersecurity and the recommendations will also vary.

If you are reading this, it means that you are interested in a career as a SOC Analyst. If you are not sure of how to begin your career and want guidance from step 1 all the way to landing your first position, click this link to register for my affordable hands-on training: [Career UP SOC Analyst Academy](#)

My training program is focused on giving you the skills and experience employers are looking for through hands-on training.

My hope is that you find some useful information within this guide to help you on your journey into becoming a SOC Analyst.

Some people will find their way on their own, others will attend school, and for others the Career UP SOC Analyst Academy will help them begin their careers.

What is important is for everyone to realize that you **CAN** have a career in Cybersecurity as a SOC Analyst. Remember, everyone began from ZERO, the ones who succeeded and have the high-paying careers are the ones who took action to begin their career. They did it, I did it, and so can you!

Learning, Dedication, Creativity, and Resilience are just a few traits that will help you along the way.

What are the 7-Steps to your SOC Analyst Career?

Step 1 - **Finding** your SOC Analyst **interest**

Step 2 - Learn about **SOC Analyst** positions

Step 3 - What experience and skills do you **have**?
What to you **need**?

Step 4 - Make your **plan**!

Step 5 - Fine tune your **knowledge**!

Step 6 - Creating a GREAT **Resume**

Step 7 - **Insights** into the SOC Analyst position

BONUS: Introduction to **Cybersecurity**

Step 1 - **Finding** your SOC Analyst *interest*

The very first step you need to take when you begin a career as a SOC Analyst is to answer the question: Why do I want to become a SOC Analyst?

Cybersecurity is an ever evolving field, attacks constantly change.

Malicious users use different attack vectors, techniques, tactics, methodologies, tools, and even reuse existing attacks to expand attacks into new hardware and software.

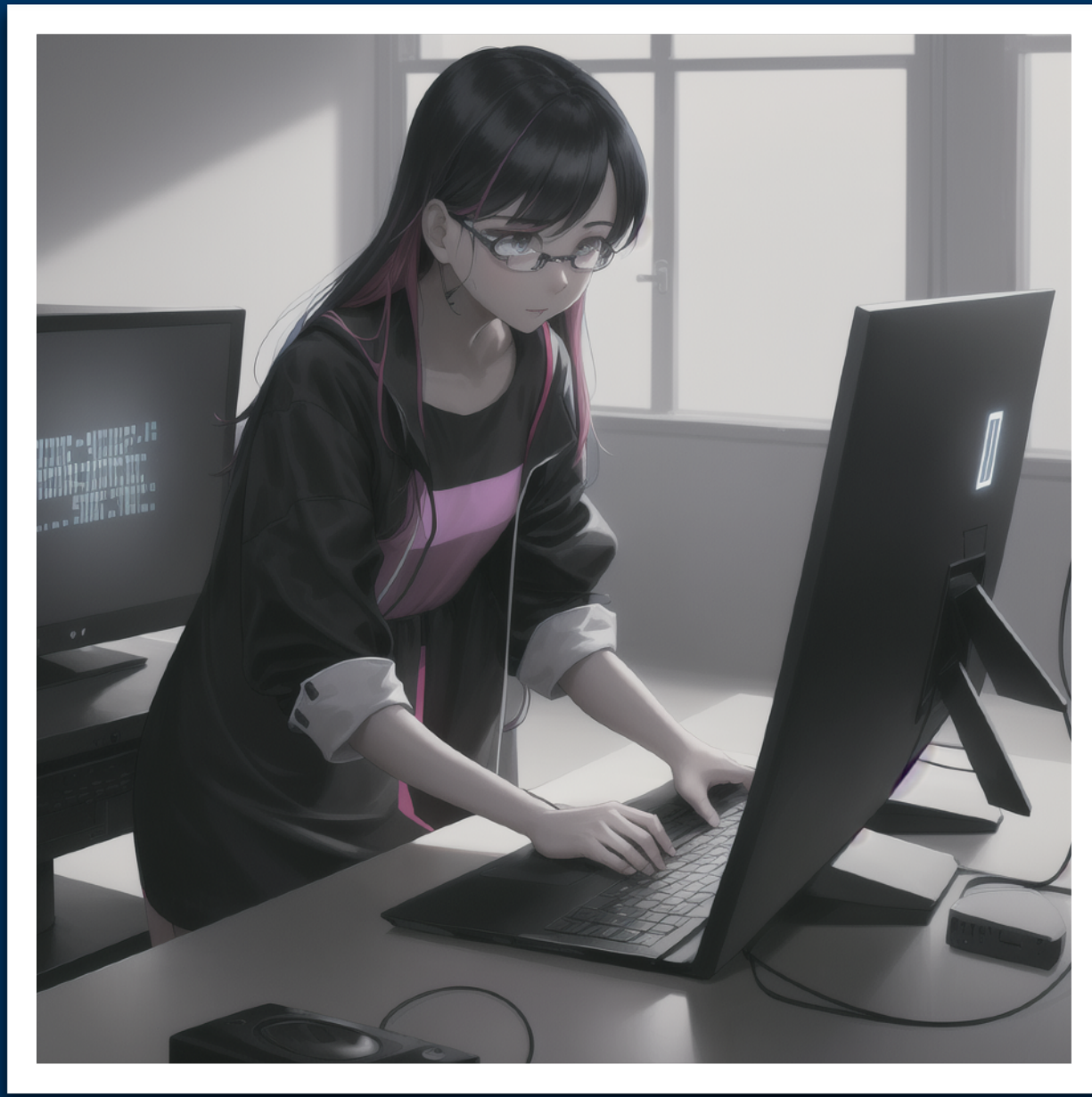
They constantly find new ways to exploit existing vulnerabilities.

Often, they are highly skilled, at times unskilled, however, they are always motivated.

SOC Analysts are at the forefront and are responsible for identifying, alerting, and stopping attacks.

SOC Analysts always in demand even in a slow economy. The opportunities are increasing and the pay can start from 30K and go beyond 100K per year.

Step 1 - **Finding** your SOC Analyst *interest*



Do you like working alone or with a team?

SOC Analysts often work in a team, with a partner, or remotely.

SOC Analysts use a variety of tools to monitor the network and systems to detect attacks.

Companies rely on SOC Analysts to keep their systems secure and to take action when attacks happen.

When attacks happen, SOC Analysts are the first to react by alerting and taking actions to stop the attack.

After the attack is under control, the SOC Analysts often help create signatures to detect attacks for faster analysis.

SOC Analysts are Cybersecurity detectives who analyze data to find security issues on networks and systems.

Step 1 - **Finding** your SOC Analyst **interest**

SOC Analysts range from entry-level all the way up to Senior levels. SOC Analysts require a variety of skills to protect networks and systems.

One of the most common skills that we will discuss below is - **Analysis**.

Analysis - SOC Analysts work with multiple tools to review characteristics, trends, and analytics to evaluate security events. Analysis is used for evaluating network and system data that is collected to determine outcomes.

TIP: Learning a programming language will be incredibly useful. If you don't know how to program, check out my Python for complete beginners course that has taught hundreds how to program with confidence.

Click here: [**Learn Python With Chris**](#)

SOC Analysts often write different scripts to review logs and also write signatures to detect attacks. Learning a programming language can help you adapt and learn quickly.

Also, employers highly value people with programming skills.

Step 1 - **Finding** your SOC Analyst *interest*

- SOC Analysts continuously monitor network traffic and security alerts to detect suspicious activities or potential breaches.
- They analyze and investigate security incidents, identifying the nature and scope of the threat.
- SOC Analysts use various tools such as IDS, IPS, Security Information and Event Management (SIEM) systems to correlate and analyze data from multiple sources.
- They work to classify incidents, determine their severity to take the necessary next steps.
- SOC Analysts collaborate with incident response teams to contain, mitigate, and recover from security incidents.
- They contribute to the development and improvement of security policies, procedures, and incident response plans.
- SOC Analysts need to stay updated on emerging threats, vulnerabilities, and security best practices.
- They report their findings and actions to management to enhance the organization's overall cybersecurity posture.

Step 1 - **Finding** your SOC Analyst *interest*

SOC Analysts need to understand how network traffic flows along with how protocols work.

Understanding how Operating Systems such as Windows and Linux work are also important. Knowledge of how attacks can affect applications, services, and systems are vital to becoming a SOC Analyst.

As a SOC Analyst, you will constantly need to learn because new attacks happen frequently, often evolving every second/minute, and sometimes even faster!

You will need to constantly sharpen your knowledge to keep up with the changes.



Step 2 - Learn about **SOC Analyst** positions

Now that you have an understanding of what a SOC Analyst and what they do, it's time to begin to learn the specifics of what SOC Analysts need to know.

It's time to explore the different areas, tools, and skills you will need to be a successful SOC Analyst.

Some of the common requirements are knowledge of:

Cybersecurity Fundamentals

IDS - Intrusion Detection Systems

IPS - Intrusion Prevention Systems

SIEM - Security Incident and Events Managers

IR - Incident Response

Malware and Anti-Virus Endpoint Security

Network Security

Operating Systems analysis

Vulnerability Assessments

...and many others



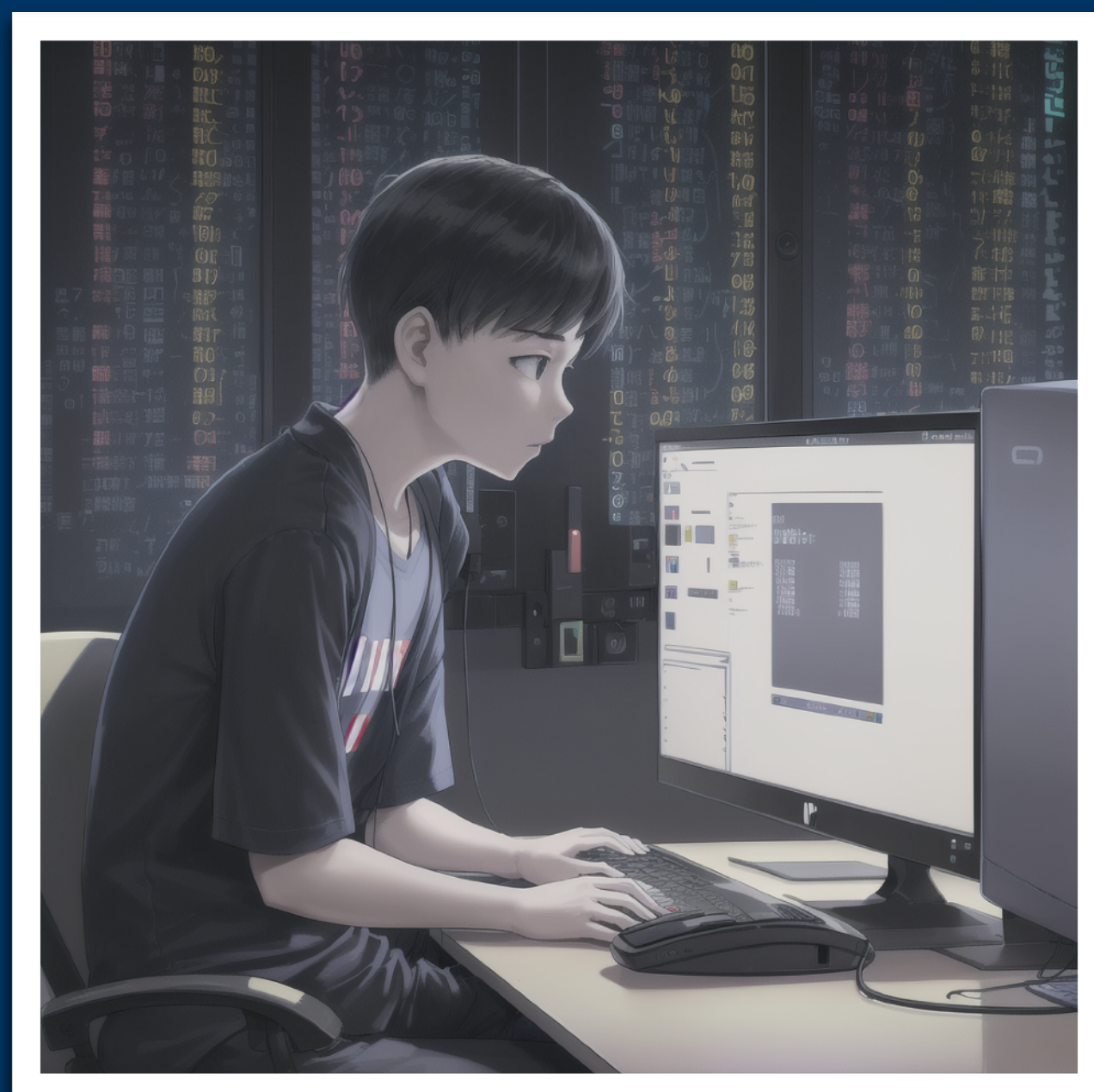
Step 2 - Learn about SOC Analyst positions

I recommend searching for SOC Analyst positions using various job sites to learn the requirements, responsibilities, and experience.

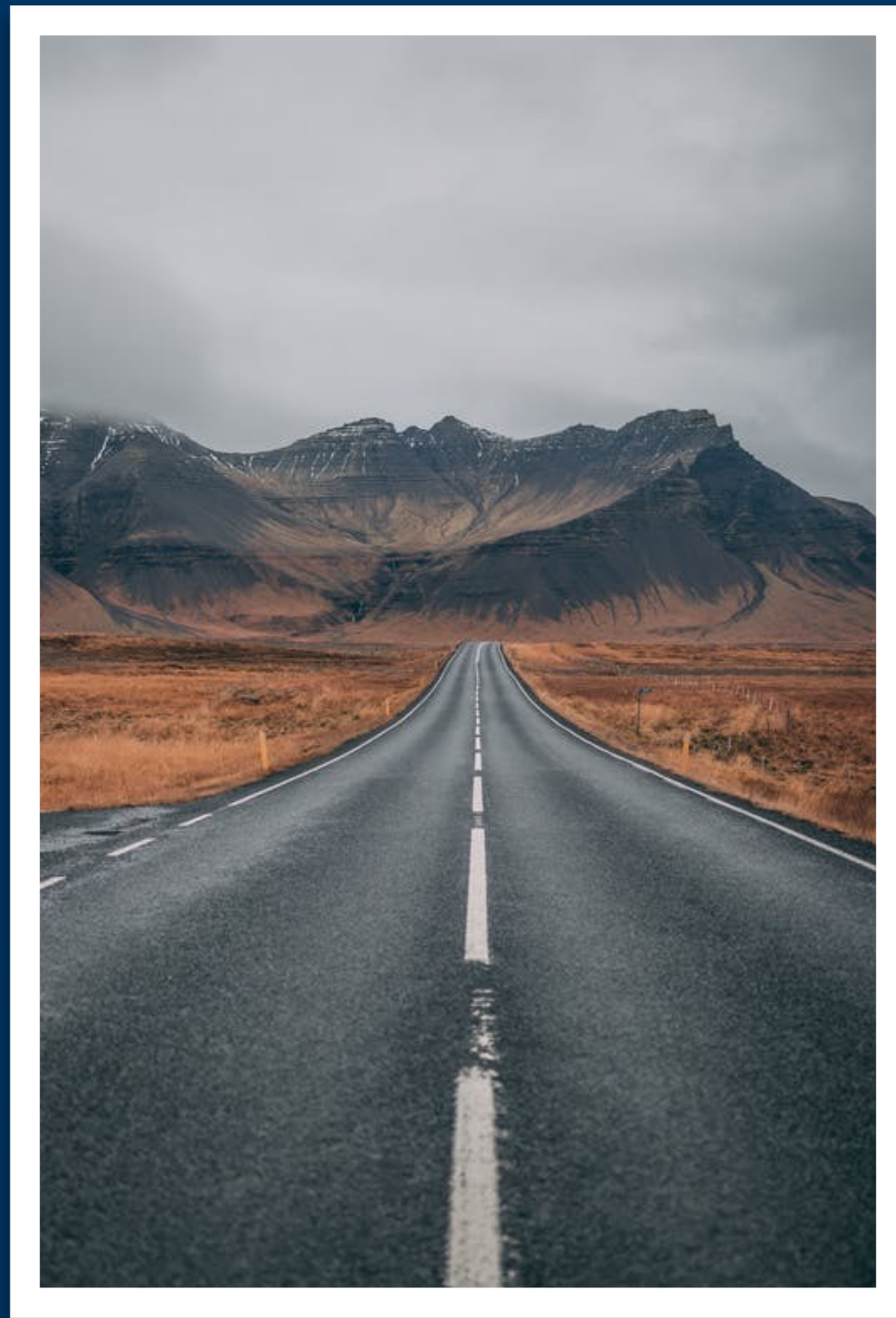
How? The easiest way to obtain the data is through your favorite job site: indeed, glassdoor, zip recruiter, simplyhired, and of course any position listings on company career sites. You can also use Google to search for: SOC Analyst job requirements.

I recommend reviewing five or more positions to allow you to begin to gather information that is commonly found within the job descriptions. You should begin to see the common requirements.

Of course, depending on where you live, the requirements may differ along with the pay.



Step 2 - Learn about SOC Analyst positions



You are now one step closer, and have better insight into what employers requirements are for a SOC Analyst.

You will also begin to learn what the most common brand of tools are used since most employers will list the brands of software and hardware used.

Starting your career is the result of having a good plan.

Now, allow me to provide you some industry insight from my 20 years of working in Cybersecurity.

Often, the requirements will appear to be daunting and you will see words and descriptions which make you think that you need a doctoral degree for an entry-level position! Here is where you really need to focus in on what I call the “**core**” functions of the role.

Step 3 - What experience and skills do you **have**? What to you **need**?

Let's quickly recap. We learned what a SOC Analyst does. We then performed some data collection for the position's requirements.

Now, it is time for you to determine what knowledge, skills, and experience you currently have.

If you are new to Cybersecurity and the role of a SOC Analyst, don't worry.

I have helped many people begin their careers where they've had **NO EXPERIENCE**, some without IT experience as well. Let's first set your mindset: **You can do this!**

In this step, it's time to write down what IT and Cybersecurity skills you have. Believe it or not, if you've touched a computer, you have experienced some levels of Cybersecurity. Locking your phone, setting up accounts such as your email, Facebook, or any other account which required some security is experience.

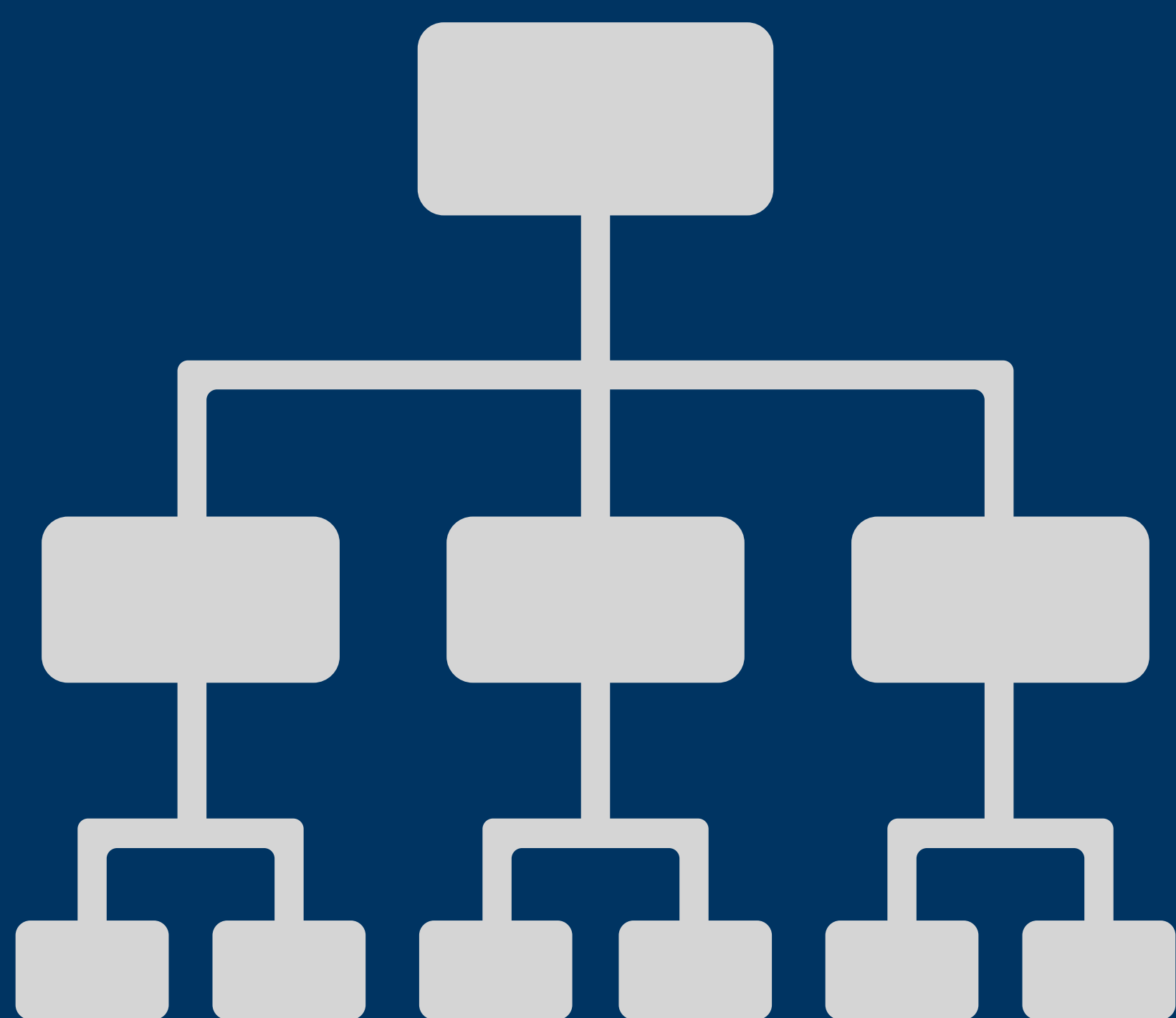
Step 3 - What experience and skills do you **have**? What to you **need**?

You may have worked in an IT role or have some experience with IT. Perhaps your position had nothing to do with what SOC Analyst does.

However, I can assure you that the chances are, you were applying some manner of Cybersecurity practice, which helps when it comes to becoming a SOC Analyst.

If you have never worked an IT job, that’s fine, we all have to start somewhere right?

In this step you will evaluate your **technical skills** to understand your current knowledge, skills, and experience to figure out your next steps. You may have knowledge that can transfer and help you become a SOC Analyst.



Step 3 - What experience and skills do you **have**? What to you **need**?

Create a list of your qualifications and skills then see if they match directly or indirectly with any with the requirements for the SOC Analyst positions you researched in Step 2.

Here is where you want to be **honest**. You will use these to build your resume and state your experience.

Here's another tip: ***If you are good with any aspect of Cybersecurity such as problem-solving, programming, math, learning, or any other type of related skill, include that in your list of skills!***

The SOC Analyst position is **dynamic**. Employers look for are people who match or have the potential to perform the job. Often, this can come through direct experience, or indirect such as people who show that they are good with analysis, learning, or problem-solving.

If you have certifications, school, or any related experience, highlight them on your resume. If you do not, don't worry we'll figure out what you need...or don't need!

Your goal in this step is to simply begin to list out your qualifications and skills, it will give you insight into what you have or what you need to apply for positions.

Step 4 - Make your **plan!**



It's time to make your plan. Now that you have compared your current knowledge, skills, and experience to what is needed for a SOC Analyst position, the next step is to make your plan.

By now, you should have a view of how your qualifications and skills compare to the position's requirements.

Your plan should be based off your experience, qualifications, and skills.

The data you collected will give you insight into what employers are looking for and how your qualifications do or do not match up to the SOC Analyst roles you reviewed.

Step 4 - Make your plan!

If you find that your experience, qualifications, and skills match up well, that is great! The next step will be to make a plan to land your SOC Analyst position.

If you find that you do not have the experience, qualifications, or skills that's ok too!

Click this link to learn how the Career UP SOC Analyst Academy can help you.

[Career UP SOC Analyst Academy](#)

Here is where we make your plan.

If your experience, qualifications, and skills aligns with a position, use the following plan as a checklist to ensure that you are not missing any elements which may work against you.

If you are starting from the beginning, let's build your plan to help you land your first SOC Analyst position.

Step 4 - Make your **plan**!

The **plan** is to step you through a sequence to establish experience, qualifications, and skills.

Establish experience that relates to the role you want to fill. This can be done by either looking for opportunities around you that allow you to apply Analysis or Engineering to help build experience you can show on your resume.

If your current job or experience is related to IT, look for opportunities to involve yourself in any SOC Analyst duties.

If you **do not** have an IT related job or do not have the opportunities, a home lab can help you build the skills you need to begin your career. This is why I include them in the SOC Analyst Academy to help you learn through real-world hands-on projects and gain the confidence to shine in interviews.

Gaining hands-on skills is the fastest way to applying what you learn and gaining the skills that employers want.

Unfortunately, reading without applying what you are learning is not always enough.

Step 4 - Make your plan!

Develop your experience. Building your own home lab and begin working on projects. Look at the position requirements and use them to work on projects that are related to the requirements.

Employers realize that not everyone has the opportunity to work in a related position, showing the employer that you have experience through projects and have applied yourself goes a long way.

TIP: Make sure you can speak to your projects in detail! Plan your projects, design, and document them.

It is important that you can detail your project to an employer and show them as much detail while illustrating that you are organized.

Why?

Often, these types of projects show the employer that you are applying areas needed to fulfill the position. Also, a lot of times, ***employers like to see someone who is detailed and driven to design and implement a project.***

Step 4 - Make your plan!

There are Facebook groups such as: [\(Link - Path to Cybersecurity\)](#), local groups ([meetup.com](#)), and even research projects that you can use to build your knowledge and experience.

VERY IMPORTANT: PROFESSIONAL NETWORKING

Create a **LinkedIn** Profile and begin to connect and build a network.

Today, it is vital that you establish relationships and a professional network.

Join groups of interest, post about what you are learning, provide information, and contribute to topics and posts.

This will help you showcase your knowledge and also begin to develop your professional network.

Also, if there are companies near you or of interest, connect with people from those companies and create meaningful conversation.

Follow and connect with me and others to learn and begin to grow your network:

<https://www.linkedin.com/in/chris-romano-career-up/>

Step 4 - Make your **plan!**

Build your qualifications by identifying the type of degree or certifications that employers are listing in the requirements section.

You will want to determine whether utilizing your degree if you have one (even if it is not directly related as nearly any B.S. degree may work) or identifying what certifications and experience are needed.

Another option is to learn a programming language. Learning a language and showcasing your programs or even being able to interview well may open up opportunities.

If you are new to IT, I cannot stress enough how helpful it is to learn a programming language. WHY? Learning a programming language gives you an advantage by teaching you how to problem solve, understand how programs operate, and learn faster. If you have never programmed or are new to IT, my course was designed for you.

You need to take the first step to begin your journey, my course can help you. [Click here](#) now, purchase, and begin learning Python.

Step 4 - Make your **plan!**

Commonly listed under the requirements for certifications, the most popular entry-level certifications are: Security+, GSEC, Splunk, GCIH, CySA+, CSA, and other vendor specific certifications.

The focus of this step is to list what employers are looking for in the position of interest. Typically, employers will list multiple certifications or degrees.

TIP: **You DO NOT need to have all of the certs or degree listed in the job position to apply!** A combination of experience and a certification or degree is usually what is needed.

Having *multiple* certifications without experience will not provide much benefit. Obtain what you need and build up your certifications as you gain experience.

That being said, I will say that ***some*** positions will REQUIRE a bachelor's degree. These are usually related to specific positions.

However, some employers will consider experience or a certification in place of a degree. This may vary depending on your position and location.

Step 4 - Make your plan!

Build your experience as you work on your certifications and or degree.

It is important that you use any opportunity to build related experience, as you work through a certification or degree. Use projects build your experience.

As I wrote earlier, use any opportunity at your current job or research and develop your own project to implement at home.

SOC Analysts monitor networks and systems, below is an example of a project you could use to help build your experience:

1. Install an attack computer: Kali Linux
2. Install a target computer: Linux/Windows
3. Install a SIEM with IDS sensors to monitor the network: Security Onion
4. Perform a write-up on the attack and how you were able to detect it.



Step 4 - Make your plan!

If you have a computer, use open-source tools, trial versions of software, or any other related material to engage and apply those skills.

Free SIEMs - Security Onion, Wazuh, GrayLog, and Splunk (evaluation)

This will help when it comes time to list out your experience and will show during interviews.

TIP: ***A degree and or a certification can work against you, if you cannot speak to the subject!*** Here is where you need to hone your skills and have confidence in the areas you have studied and applied yourself.

I can tell you from experience, the best way to learn something is to build and configure it yourself. Along the way, you may run into issues, those are the best times to really learn!

With the **GOOD** comes the **BAD**, having sound troubleshooting skills is a strong skill to display on a resume or in an interview!

Step 4 - Make your plan!

Create a schedule to complete your projects along with the documentation. As you work on your projects, organize the details so that you are able to describe them in detail.

The key to this step is to clearly walk someone through your projects in an organized manner.

Sounds easy right? Trust me, it's not as easy as it seems when you are in front of a hiring managers and perhaps a few seasoned Analysts or Engineers in an interview!

Practice by describing your project to someone who has no idea what you are working on.



Step 4 - Make your plan!

By the end, if they are able to summarize what you did in your project, then you have completed this portion successfully.

Along the way, you've begun to build up your experience, qualifications, and skills!

Also, you will find that actually working on something will help once you begin the actual position.

Will things match up exactly? Probably not however, the first time something goes wrong, your experience with troubleshooting will be helpful.

Also, you will find that you are able to gather and organize your approach to assignments. This is a skill that will benefit you from entry-level all way through the rest of your career.

Step 4 - Make your plan!

Some common SOC Analyst areas you will want to know are:

Networking basics

Operating Systems such as: Windows and Linux

Application Security basics

General understanding of protocols

Security basics of Analysis - event analysis, using dashboards, and documentation/communication/ticket creation

Offensive and Defensive Cybersecurity techniques and common tools

Programming - [Python](#), R, C++, or Bash scripting ([Click Here](#) to get started with Python today.)

Languages - SQL, HTML, and JSON

Use your position research to identify other common areas needed

Step 5 - Fine tune your knowledge!

SOC Analyst positions typically involve multiple systems. In order to protect them, you need to have a good understanding on how systems interconnect and allow users to actually use the systems.

Here is where you want to develop your well-rounded knowledge. Certifications are useful in organizing and providing material for you to learn in a particular cybersecurity area.

My recommendation, **DO NOT** simply focus on the exam itself, make sure you are able to speak to the knowledge areas of the certification!

The hiring managers and experienced Analysts are in their positions because they have solid foundational knowledge.

Understanding the different areas and how they work together is important. For example, the key areas I teach in the SOC Analyst Academy are: Operating Systems, Networking, Offensive and Defensive Security, Cloud, and Containerization.

Step 6 - Creating a GREAT Resume

You've put in a lot of hard work, now it's time to showcase it! Your resume should reflect all of your relevant experience, qualifications, and skills to show the potential employer that you are a good match.

There are many different formats to creating a resume. Here are the most important areas you will want to include when creating your resume:

Objective - Should directly identify your experience and **EXACTLY** match up to the position listed on the **position description**.

Skills and Qualifications - Detail your skills along with your certifications and education.

Work Experience - List your work history and highlight areas that align with the position. Include **exact words** that are used in the position description. Within each portion of your position description, use **metrics**, illustrate your **value**, and align the descriptions to be *relevant* to the position.

Avoid too much detail on **irrelevant** topics

Step 6 - Creating a GREAT **Resume**

Certifications - If you have certifications, list them on the first page if they directly match up to what the employer is looking for.

Education - List your education and make sure and highlight any activities, awards, accomplishments, or highlights which may be of interest.

References - Include references if required otherwise, simply state: References available upon request.

Spelling/Grammar - Use spelling and grammar checks to avoid small errors. Believe it or not managers pay attention to this!

Have someone read over your resume and provide feedback.

Step 6 - Creating a GREAT Resume

It's important to match up the keywords as resume filters may filter out a candidate who qualifies, but does not have the matching keywords Pay attention to these!

Identify the keywords and use them throughout your resume as much as you can.

The more you can match up, the better it looks and the higher the chances of your resume standing out!

Your personal information - Make sure your name and email are formal, your email should not reflect badly (avoid gamer tags or anything unprofessional)

Resume length - Your resume should be 1 to 2 pages maximum, avoid lengthy resumes!

The key to building a **GREAT** resume is matching up as much as you can in every area to the position description and requirements.

Step 6 - Creating a GREAT Resume

If you are unsure how to structure your resume, there are multiple ways you can create your resume.

Google resumes related to your position and review their format.

Pick out the ones which stand out and mimic the structure.

Use a resume builder. There are sites which provide formats that allow you to simply fill in the details and output a nicely structured resume.

Avoid the modular resumes, use the traditional or modern professional formats and follow the employers file type to avoid having your resume skipped due to a processing error.

Common formats are: docx, txt, rtf, and pdf.

Some resume building sites are free while some may charge a fee.

A hiring manager should be able to determine whether or not you are a good fit, **should be obvious in your resume immediately.**

Step 6 - Creating a GREAT Resume

Common SOC Analyst Resume Structure:

I. Contact Information

- Full Name
- Phone Number
- Email Address
- LinkedIn Profile (if applicable)
- Location (City, State)

II. Resume Summary or Objective

- A concise, compelling statement highlighting your qualifications and career goals as a SOC Analyst.

III. Education, Certifications, and Skills

- A bullet-point list of your Education, certifications, and skills relevant to the SOC Analyst role. Include both technical and soft skills.

IV. Professional Experience

Chronological order, starting with the most recent position

- **Job Title:** Start with your job title, followed by the company's name, location, and employment dates (month/year).
- **Company Description:** A brief description of the company's industry and size.
- **Responsibilities:** Detailed list of your duties and achievements in the role, focusing on responsibilities relevant to SOC analysis.
- Describe your role in incident detection, response, and mitigation.
- Mention any specific tools or technologies you used (e.g., SIEM platforms, IDS/IPS).
- Highlight any notable incidents you handled and the outcomes.
- Emphasize teamwork and collaboration with other security professionals.
- **Achievements:** Mention any accomplishments, such as reducing incident response time, improving security posture, or certifications earned while in the role.

Step 6 - Creating a GREAT Resume

TIP: **You should create a specific resume for each job you apply for!** While you do not need to recreate the entire resume, it is important that your resume directly reflects your experience, qualifications, and skills to match up as closely as possible.

This is important as using a single resume may not lead to reaching a recruiter or manager due to keyword and other resume filters.

TIP 2: **Your summary is an important area of your resume!** When you write your summary (first section) make sure that it immediately reflects your intention of filling the position and states how or why you are qualified.

BONUS TIP: Sites that are easy to apply and do not have contact information often end up with 10 times the number of applicants.

Look for a contact and follow up with an email after you apply to let them know your interest.

Also, create a LinkedIn account and connect with people at the company of interest and your area of interest. Networking is important in today's competitive job market and referrals can be an advantage for you!

Step 6 - Creating a GREAT Resume

As a hiring manager for many years, the first page is what I look at and often determines whether the candidate may be considered or not.

When I am looking for a SOC Analyst and the summary states that you are looking for something else or is not clearly stated, it does not focus your resume. This requires me to see if the rest of your experience or qualifications fit the position.

The manager should not have to determine whether or not you are a good fit, it should be obvious immediately.

Use the first page to show your interest, qualifications, and highlight areas that show that you are the right person for the position.

TIP: List the skills that you have that relate to the job first.

When you are describing what you did at previous jobs, make sure that you relate each function as closely as you can to the position you are applying to.

Step 7 - *Insights* into the SOC Analyst position

The position itself, what to expect? The first question you might have is, can I actually do this?

Did I study enough?

Did I do enough?

AM I READY?

Here's the reality, with your first position or nearly all entry-level positions, the employer is looking for someone they can teach and train.

“Most” employers are going to have you follow a training sequence or work with one of the experienced members of the Security team to help you learn HOW they work.

Step 7 - *Insights* into the SOC Analyst position

The most important aspect of the SOC Analyst position, is taking everything you have learned, and applying it in the manner in which your employer requires.

What exactly does this mean?

As you may or may not know, there are thousands of security products. Nearly every company uses what they feel works best for their need.

The key is being able to learn and apply what is needed to learn and apply the company's security methods to help protect their users, data, systems, and all other resources.

In other words, you will most-likely **not** know the job going into it!

There are some exceptions such as specific roles using specific hardware or software however, those are more specialized roles which allow you to focus as necessary.

Step 7 - *Insights* into the position

It's perfectly normal to apply for many SOC Analyst positions as you begin your career.

I want you to remember that each step good or bad is a learning process, don't become discouraged if you are passed up for a position.

Take the opportunity to ask the employer if they can give you any insight as to why you were passed up.

Sometimes they simply picked someone before you interviewed, sometimes you may need to work on an area, and sometimes it is simply not a good fit.

While difficult, do not take this personally, there are a variety of factors that are often beyond your control.

Always, continue to keep working towards your goal. If you stop, your chances are 0 - If you continue, learn, and adapt then your chances increase substantially.

Step 7 - *Insights* into the SOC Analyst position

Learning about the SOC Analyst position has several advantages, it will help you identify the skills that employers want and allow you to create a list of what you need to land the position.

Here are some details about SOC Analysts:

SOC Analysts work in a team to monitor the network, systems, and other areas for security issues or activity.

They utilize a variety of tools for monitoring including: SIEM, HIDS, IDS, IPS, DLP, Firewall and Network Activity, along with other tools.

Skill Areas:

Network Traffic Analysis

Operating Systems / Basic Coding: **Python**, Bash Scripting, MS

PowerShell Incident Response

Report Creation

Tool configuration

User support

Communication

You may like this if:

You like working with a team, users, and communicating Monitoring and Analyzing Network Traffic and Events Investigating Events

Incident Response

Report Creation - Basic Statistics Configuring Security Tools

Research Attacks and Trends

If you are interested in becoming a SOC Analyst and want to learn with real-world fundamentals and hands-on projects.

Visit: <https://www.careerup.tech> to learn how you can launch your career in less than 3 months through the Career UP SOC Analyst Academy.

Introduction to Cybersecurity

Security Breaches - A Security Breach is when the security controls that enforce Confidentiality, Integrity, and Availability (CIA Triad) are compromised.

It is important that you understand that security breaches can be caused by internal users by accident, neglect, or on purpose. In addition, they can be caused in the same manner by external users. Of course, we have malicious users who intend on compromising all or some of the CIA triad.

The function of a security team is to enforce and ensure that mitigations are on place to protect systems. The overall function of the security team is to uphold all areas of the CIA.

CIA: Confidentiality, Integrity, and Availability:

Confidentiality - Maintaining data such that it is only accessed by intended audiences.

Integrity - Maintaining that data is preserved in its original, unaltered state as intended.

Availability - Providing the services and functions to maintain access to resources, systems, and other needs as intended.

The opposite of **CIA** is **DAD** or Disclosure, Alteration, and Denial:

Confidentiality - **D**isclosure

Integrity - **A**lteration

Availability - **D**enial

Introduction to Cybersecurity

CIA VS DAD

Think of these as “Defensive” VS “Offensive” purposes.

Where a security team preserves the CIA triad, malicious actors are motivated to carry out the functions of DAD to disrupt or compromise the security posture of an organization.

CIA and DAD were developed to provide a starting point to define security functions. Think of both of these as the starting point of any security program, or framework.

WHY?

Consider the issues faced if an organization suffers a breach or breakdown in one of their security areas:

Breach Impact - The Breach Impact defines the damage caused by an incident. These include:

Financial Risk - This is the amount of money lost due to an incident.

Reputational Risk - The amount of trust loss, branding damage, and of course reputational damage. For example, if a bank is compromised and your account is affected, would you trust the bank?

In addition, a side-effect is “Personal Identifiable Information” PII compromise, breaches can allow malicious actors access to your information. Birthdate, Account Number, Social Security Number, Address, Password, Password hints, PIN, and other information that could be used not only to compromise the current account, but other accounts as well.

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Strategic Risk - This risk involves the disruption of critical tasks, processes, operations, or any other required function that prevents an organization from meeting goals or objectives. For example, suppose a car factory's robotics systems are compromised and cannot piece together a portion of a car.

Another example is a shampoo manufacturer that has a database compromised, which does not have a backup, and now the ingredient mixtures are unknown. Notice in these scenarios that the “strategic” risk involves disruption of an organization to carry out their primary goals or objectives.

Operational Risk - The disruption of carrying out normal day-to-day operations. For instance, attacking the payment processing system of a store may require manual entry to check customers out. The operations proceed however, they are impacted, which affect the company.

Think of this as disruption of normal activities ie: “Operations” that cause an organization to alter the normal ways of working or completing functions.

Compliance Risk - This risk involves the failure to meet compliance, for example, a hospital not safe-guarding medical records. Another example is a bank not securely processing transactions. Both of these compliance risks not only pose informational risk, they violate compliance, therefore posing risk to the users who depend on the compliance to safe-guard their information.

This in turn violates regulation, which has consequences, fines, and may even halt operation if the violation is severe.

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Implementing Security Controls

There are **three** types of Security Control Categories you will need to know:

1. **Technical Controls** - The “technical” or think of any electronic/digital control which is used to enforce the CIA triad.

Examples, Firewalls, Access-Control Lists, Intrusion Prevention Systems (IPS), Implementing Operating System controls (Windows: Group Policy, Registry, Windows Firewall) or Linux: Service/File restrictions. Security Enhanced Linux (SE Linux)

2. **Operational Controls** - These are the processes and procedures set in place to manage security and define the manner in which they are taken.

3. **Managerial Controls** - These focus on the procedures to manage risk such as: Risk Assessment, Change Management, Configuration Management, Project Management, and other “management” activities which oversee and define how risk is managed.

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Consider how each of these applies to the CIA triad:
Some examples:

***Technical* - Confidentiality** - Limiting permissions on files, folders, services, accounts.

***Operational* - Integrity** - Writing a procedure that requires any files that are copied to media to be encrypted.

***Managerial* - Availability** - Performing a risk assessment to determine the level of redundancy and impact of system operations. Determining what effects a system has on functionality if it fails.

Additional Control Types:

Preventative Controls - They prevent an event from happening, example: Firewall blocking traffic

Detective Controls - They identify when an event occurs, example: Intrusion Detection System (IDS)

Corrective Controls - These correct issues, example: restoring a system after malware infection from an image.

Deterrent Controls - These deter an attacker, for example: A banner on a system which states that any intrusion attempts are recorded and will be reported to authorities for prosecution.

Physical Controls - These are physical and prevent “physical access,” examples: Fence, locked door, gates, human guards, barbed wire fence, biometric access readers, anything physical.

Compensating Controls - These reduce or eliminate risk due to exceptions to a security policy, examples: ***least privilege, zero trust***

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Protecting and Securing Data

Data is considered one of the most valuable assets today. Data can contain valuable information, therefore malicious actors tend to target data.

There are **three** states of data where security must be considered:

Data at rest - This is data that resides on media such as hard drives, USBs, or any other media that stores data.

Data in motion - This is data transfer such as data traversing a network. Think of this as when you send your password or any sensitive data, as it traverses a network, how secure is it? Example: HTTP compared to HTTPS where HTTP is cleartext, meaning anyone intercepting the data, can read it! HTTPS encrypts data via TLS which uses public key infrastructure which uses certificates to validate authenticity.

Data in processing - This is data that is being used by a system. For example, a database transaction, as records are updated, they are in use. The system must adhere to securing these processes. Often systems use memory isolation along with process isolation to uphold integrity during processing.

Think of this as user1 is performing a database update at the same time as User2. However, the system will separate and isolate both processes to prevent User1 or User2 from compromising integrity of either operation.

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Data Encryption - Data that someone can read, which is unprotected is “Vulnerable” therefore, encryption is used to transform clear-text data ie: data we can read, such as what you are reading right now...into cipher text, which is data that is “encrypted” via algorithms which use complex mathematical operations to obfuscate or “confuse” or “diffuse” data into an unreadable format.

This preserves...you guessed it....”Confidentiality” since only the intended reader can “unencrypt” or revert the data back into “human readable” format.

Data Loss Prevention (DLP) - Data loss often occurs when data is leaving or removed from a secured system or network. Therefore “DLP” is used to monitor for specific data leaving a system or network by detecting specific data via keywords, patterns, watermarking, hashing, or policy. This helps prevent data theft and adds in a control to help mitigate data loss via intentional or unintentional transmission.

There are two types of DLP commonly used:

Host-Based DLP - Hosts are monitored for data transmissions and policies alert and in some cases restrict transmissions.

Network-Based DLP - Networks are monitored to detect and prevent data from leaving a network.

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Minimizing Data Risk

Data often has sensitivity, some data may even be classified by importance, restriction, or even a security label.

Depending on the data, it may be in the best interest to minimize access or even the existence of the data if it is not in use.

Depending on where the data is stored, it may be destroyed (physically) or transformed into a format, which no longer represents the data. Another process is to remove the known source or link of origin, this is known as de-identification.

Another alternative is **data obfuscation**, this essentially transforms the data, in a manner that no longer represents the original.

There are two methods:

1. **Hashing**, or using a one-way algorithm to transform the data from its *original* to a *hash* or transformed output.
2. **Tokenization**, where a sensitive value is replaced, and is only known via the lookup table. This involves, replacing the actual data with randomized data, to avoid the original data from being known. For example, replacing your social security number with randomized numbers or characters, to avoid anyone knowing your actual social security number. ONLY the values in the lookup table, will match the randomized characters.

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What is a **Cybersecurity Threat**?

This is a common question and one that you may be asked in an interview.

Well, first let's discuss "**Cybersecurity**" - What exactly is Cybersecurity? (You would be surprised how many people in the field, with or without degrees...who cannot answer this question)

...so let's begin by defining Cybersecurity:

Cybersecurity is the area within Information Technology that specifies the processes, procedures, and techniques for protecting technology from intentional and unintentional threats and vulnerabilities.

Next, let's define a "**Threat**" - Simply put, a Threat is any potential occurrence of unintended consequence which adversely impacts an organization's operations, assets, individuals, or related entities.

There are multiple different definitions, and they tend to change as the attack surface and technologies change.

Remember, Cybersecurity constantly evolves, this is why I focus on providing you with the "thought processes" to think as a Cybersecurity professional, rather than focusing on any specific area or technology (trust me on this one!).

So, what is a **Cybersecurity Threat**? It is the potential of an intended or unintended occurrence where the threat adversely compromises the Confidentiality, Integrity, or Availability in an organization. Often, you will hear a Cybersecurity Threat as something specific such as virus, malware, technique, or as a "Threat Actor"

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Last but not least!

Open Systems Interconnection (OSI) 7-Layer Model

The OSI model was created to help vendors standardize how software and hardware communicate. In the earlier days of the IT, vendors required specific equipment in order to communicate. This became problematic hence, standards were developed to allow vendors to work with each other. Below is an overview of the OSI 7-Layer model. I highly recommend knowing this model before any interviews!

OSI Layer	Function
Layer 7: Application	Application interaction from humans takes place
Layer 6: Presentation	Data formats: Text, Pictures, Movies, Audio, Encryption, and Compression (all data formats)
Layer 5: Session	Sessions are created, authentication, permissions, and session ended
Layer 4: Transport	Data transmissions are controlled through communications - PDU: Segment
Layer 3: Network	Physical Paths are decided, logical addressing, and routing - PDU: Packet
Layer 2: Data Link	Physical Addressing, switching, logical link control, MAC addresses, VLANs - PDU: Frames
Layer 1: Physical	Physical interface - actual physical equipment - PDU: Bits

Congratulations! Now it's your turn...

This was an introduction to Cybersecurity, if you are interested in developing the hands-on skills and learning even more about these topics and more, click the links below to enroll my Cybersecurity bootcamp.

Whatever you do, never stop! I went from stocking shelves to signing 20+ million dollar projects, it wasn't always easy, but I never quit. I did it and so can you!

Click the following link go get started today:

[Career UP SOC Analyst Academy](#)

Don't fall behind, click to learn Python Programming today:

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Now, go start your career!

Thank you,
Chris

[Follow me on LinkedIn](#)